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Computer Vision
and Learning
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Advancing (Sparse) Multi-View Depth Estimation

Bachelor's Thesis

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Abstract

Zusammenfassung

Acknowledgements

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1

Introduction

Recovering the three-dimensional structure of a scene from visual observations is a foundational problem in computer vision with critical applications in robotics and autonomous navigation [2]. While spatial-first foundation models such as MAST3R [4] and PI3X [7] have achieved remarkable per-frame geometric fidelity, their application to 4D sequences - where temporal coherence is as vital as spatial accuracy - remains an open challenge. In sparse-view configurations (2-4 cameras), the problem is bifurcated: spatial models lack the temporal grounding to prevent high-frequency jitter, whereas dedicated 4D models, such as VGGT4D [3], introduce temporal priors that risk over-smoothing geometry or creating "motion-blur" artifacts.

This thesis investigates the transition from per-frame 3D reconstruction to sequence-level 4D modeling. We evaluate whether the current generation of foundation models can maintain metric integrity across time or if the introduction of temporal constraints inevitably degrades geometric precision.

1.1 Background and Motivation

Classical multi-view reconstruction methods such as SfM [5] provide high rigour but struggle with the textureless regions and dynamic occlusions inherent in human-centric environments. Learning-based spatial models like DUST3R [6] and PI3 [7] have largely superseded these by regressing dense pointmaps directly, yet they treat video as a collection of independent snapshots. This independence results in significant temporal instability when

these models are deployed on robotic platforms requiring smooth trajectory estimation.

To address this, temporal-first architectures like VGGT4D [3] and MonoFusion [8] incorporate motion cues to encourage sequence-level consistency. However, the nature of this consistency - whether it acts as a beneficial geometric anchor or a destructive low-pass filter - is poorly understood in sparse-view settings. There is a critical need for a benchmarking framework that quantifies the "temporal cost" of these various architectural choices.

1.2 Problem Statement and Contributions

The central objective of this work is to characterize the performance of 3D and 4D foundation models under the strict hardware constraints of sparse-view (2–4 camera) setups. We focus on measuring the divergence between per-frame accuracy and sequence-level stability, identifying where hierarchical vs. graph-based alignment strategies succeed in mitigating drift.

The main contributions of this work are:

- **4D Benchmarking Framework.** A unified evaluation pipeline for sparse multi-view video that subjects six base models (MAST3R-SfM, VGGT, PI3, PI3X, VGGT4D, and MonoFusion) to identical data-loading and semantic masking conditions across the DEX-YCB, Hi4D, and 4D-DRESS datasets. Crucially, this framework evaluates the impact of applying the GGPT [1] refinement strategy across these architectures (excluding MonoFusion), providing a direct comparison between raw spatial outputs and geometry-grounded 4D sequences.
- **The Delta-Consistency Metric.** We formalize a novel quantitative measure to isolate the "temporal cost" of sequence alignment. This metric, $\Delta_{\text{consistency}}$ (see Equation (1.1)), captures the divergence in geometric fidelity when per-frame reconstructions are transformed from independent, local coordinate systems into a globally consistent 3D reference frame. This value exposes the spatial degradation caused by alignment drift and temporal constraints.

$$\Delta_{\text{consistency}} = \text{Chamfer}_{4\text{D}} - \text{Chamfer}_{3\text{D}} \quad (1.1)$$

- **Comparative Alignment Study.** A rigorous analysis of three temporal alignment strategies, highlighting a performance split: identifying

hierarchical binary-tree methods as superior for static scene completeness and graph-based optimization (PGO) as superior for dynamic accuracy and minimizing $\Delta_{\text{consistency}}$.

This section must include explicit references to the detailed method descriptions in later chapters (hierarchical alignment trees + PGO formulation).

- **Staticity-Informed Refinement.** An investigation into refinement strategies - such as staticity constraints applied to MAST3R-SfM - aimed at reducing high-frequency jitter without the over-smoothing seen in purely temporal architectures.

2

Background Knowledge

3

Related Work

4

Methodology

5

Experiments

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Limitations

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Future Work

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Conclusions

Bibliography

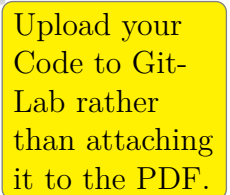
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List of Acronyms



Artifacts Availability

The artifacts associated with this thesis can be found in the ifi-GitLab¹. Within this repository, there are two main folders: one containing the code to fine-tune the model, and the other containing the code to generate the training data, including all the different datasets. The README file provided in the repository contains step-by-step instructions on how to set up the environment used in this thesis, install all necessary dependencies, and run the source code.



Upload your Code to Git-Lab rather than attaching it to the PDF.

¹<https://gitlab.ifi.uzh.ch/ddis/Students/Theses/<NAME>>



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Zürich, April 27, 2026

Signature of
Fabio Di Meo